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Machine learning framework for the analysis and prediction of energy loss for non-fullerene organic solar cells

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ABSTRACT

The efficiency of organic solar cells (OSCs) has been improved more than 19% recently with the development of non-fullerene acceptor materials. Further improvement is still attainable with the optimal combinations of donors and acceptors that provide minimal energy losses. In this work, a data-enabled machine-learning (ML) framework was employed to predict the energy losses in the polymer:non-fullerene acceptor based devices. Based on the collected experimental dataset, the prediction accuracies of various machine learning models were systematically compared by estimating mean absolute percentage errors (MAPE), root mean squared errors (RMSE), and person's r coefficient. The Random Forest regression model showed the best performance in predicting the energy losses with a correlation coefficient of 0.83 and relative error in the range of 0 – 20%. The predictive ability of this model was further validated using the different parameters of devices with power conversion efficiency range of 6 – 18%. Three different donor–acceptor combinations were chosen for fabricating the photovoltaic devices to fit this model into practical devices and experimentally obtained energy loss values were compared with the predicted values. In addition, the device parameters with the molecular descriptors to understand the correlation and energy loss is highly correlated with the HOMO offset. This study demonstrates that the ML approach provide an effective method to predict and virtual screen of promising donor–acceptor pairs with minimal energy loss and would be useful for developing next-generation high performance solar cell materials.

1. Introduction

In recent decades, organic solar cell (OSC) has emerged as one of the most promising photovoltaic technologies due to its advantages like solution-based manufacturing process, lightweight, mechanical flexibility and low energy payback time (Di Carlo Rasi and Janssen, 2019; Han et al., 2021; Heeger, 2014; Lu et al., 2015). However, the commercialization of OSCs is delayed due to their high energy loss and low environmental stability (Agrawal et al., 2016; Duan and Uddin, 2020; Jørgensen et al., 2012). To achieve high performance devices with good stability, novel D-A conjugated polymer, small molecule donor and A-DA'D-A based non-fullerene acceptor (NFA) materials with promising properties and various novel device architectures have been developed (Karak et al., 2018, 2015; Keshtov et al., 2020; Liu et al., 2021; Sharma et al., 2021; Zheng et al., 2020). Among these materials, NFAs have achieved widespread research attention due to their tunable energy levels, high optical absorption in NIR range and excellent charge transport properties (Karuthedath et al., 2021; Wu et al., 2020; Zhang et al., 2018). By utilizing these properties, high open circuit voltage (V_{OC}) and short circuit current density (J_{SC}) have been simultaneously achieved (Li et al., 2021; Liu et al., 2020; Liu et al., 2020; Sun et al., 2021; Yuan et al., 2019). Recently, Sun et al. attained a record power conversion efficiency (PCE) of 18.37 % in NFA based binary bulk heterojunction device (Sun et al., 2021). However, the performance of OSCs still lags behind that of silicon and perovskite solar cells due to their higher energy losses (Almora et al., 2021a, 2021b; Green et al., 2022). Previous theoretical studies based on Shockley-Queisser (SQ) limit for OSCs indicate that more than 20 % PCE can be achieved if the energy loss can be minimized below 0.4 eV for such material systems(Hou et al., 2018; Sandberg and Armin, 2021; Upama et al., 2020). The energy losses in OSCs largely depend on the charge transfer (CT) between the donor and acceptor materials and the radiative and non-radiative recombination of charge carriers (Benduhn et al., 2017; Hofinger et al., 2021; Ji et al., 2020; Sharma et al., 2020). It was observed that the optimal offset values between donor and acceptor materials can significantly reduce the losses (H. Liu et al., 2021; Yuan et al., 2020).

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Therefore, it is essential to understand the correlation between molecular parameters and energy losses to propose new strategies that will accelerate the development of high-performance stable devices for their successful commercialization.

To achieve highly efficient devices with minimal energy losses, the ideal donor–acceptor (D-A) material combination must be identified. However, this selection procedure by carrying out trial and error experiments is often time consuming and expensive. The data-driven machine learning (ML) approach is one of the efficient and straightforward techniques that enables identifying the materials with suitable properties (Butler et al., 2018; Mahmood and Wang, 2021; Yılmaz and Yıldırım, 2021). Moreover, this approach can provide better understanding on the relation between the material properties and performance parameters of the devices. In recent years, the researchers have explored various data driven ML approaches to predict device efficiency, photocurrent, stability and non-radiative voltage loss, etc. (David et al., 2020; Lee, 2020; Malhotra et al., 2021; Rodríguez-Martínez et al., 2021b; Sahu et al., 2018). Sahu et al. developed the ML model to predict PCE using 13 macroscopic properties of organic materials and this model showed impressive performance with Pearson's coefficient (r) of 0.79 (Sahu et al., 2018). Lee et al. developed an ML model based on random forest (RF) regression algorithms to predict the PCE using electronic descriptors such as molecular orbital energy levels and band gap of donors and acceptors (Lee, 2020). This model provided high predictive power with a coefficient of determination of 0.80. Recently, Malhotra et al. established a correlation between non-radiative voltage loss and the molecular and structural descriptors using different ML model, and predicted the non-radiative voltage loss with r = 0.85 (Malhotra et al., 2021). Furthermore, different ML techniques have been developed for screening and designing of the suitable active layer materials (Kaka et al., 2022; Kranthiraja and Saeki, 2021; Lee, 2022; Richards and Paul, 2021; Rodríguez-Martínez et al., 2021a; Sahu et al., 2019; Sun et al., 2019; Y. Wu et al., 2020). Although different ML techniques have been effectively employed to investigate the structure-performance relationship in OSCs, an ML approach to predict the energy loss using experimental oriented molecular descriptors has not yet been realized. In this work, ML algorithms were developed for predicting the energy loss in polymer:NFA active layer based OSCs. However, the devices based on all polymers or small molecules, ternary blends and tandem structures are not in the focus of the current study.

Herein, a systematic ML approach was carried out using readily available molecular descriptors such as energy levels and offsets, band gap, different processing conditions, etc. For this purpose, five different ML models such as decision tree (DT), support vector machine (SVM), gradient boosting (GB), random forest and artificial neural network (ANN) were trained. All these models performed reasonably well on the collected dataset. A statistical analysis on the same dataset was also carried out to capture the major trends in the NFA based OSCs. The real OSCs were also fabricated with three different photoactive layers (PBDB-T:ITIC, PM6:Y6 and PM6:BTP-eC9) to validate the obtained ML results, and these devices exhibited comparable values in predicted and experimental energy losses. The current study showed a steady improvement in efficiency of NFA based devices by decreasing the total energy loss. In this perspective, a new strategy to attain high performance devices can be proposed for minimizing the energy offset and recombination losses in novel conjugated polymer and NFA materials. This report will also provide valuable information to better understand the influence of macroscopic properties of active layer materials on the performance of devices.

2. Energy losses

The energy losses in OSCs are mainly originated from the non-radiative charge recombination and energy offsets at the donor–acceptor interface. The expression, $E_{loss} = E_g - qV_{oc}$ defines the energy loss in OSCs in terms of bandgap (E_g) and V_{OC}, where E_g denotes the optical bandgap of active layer and q is the elementary charge (Yao et al., 2015). Based on the detailed balance theory, the total energy loss can also be expressed as the sum of three different kinds of measurable or known losses as shown in equation (1) (Wang et al., 2018; Yao et al., 2015).

$$E_{loss} = (E_g - qV_{OC}^{SQ}) + (qV_{OC}^{SQ} - qV_{OC}^{Rad}) + (qV_{OC}^{Rad} - qV_{OC}) \tag{1}$$

$$E_{loss} = q\Delta V_1 + q\Delta V_2 + q\Delta V_3 \tag{2}$$

where V_{OC}^{SQ} and qV_{OC}^{Rad} are the maximum V_{OC} in SQ limit and radiative limit for a particular E_g, respectively. The schematic diagram representing the different types of energy losses are shown in the Fig. 1. The first term of the equations (1) and (2) indicates the energy loss induced due to the mismatch of solar irradiation in a solid angle and omnidirectional blackbody radiation of the absorption above the E_g. This energy loss ($q\Delta V_1 \sim 250$–300 meV) is inevitable and it depends only on the bandgap of the materials (Hou et al., 2018). The second term represents radiative energy loss induced by the photon absorption that occurred below the band gap, which is relatively low ($q\Delta V_2 \leq 50$ meV) in the case of efficient devices with low energy offsets (Yuan et al., 2020). The non-radiative energy loss ($q\Delta V_3$) induced due to the presence of defect and impurity states is represented by the third term of the equations (1) and (2). For highly efficient OSCs, this energy loss $q\Delta V_3$ is in the range of 200–300 meV (Wang et al., 2021). Based on the data collected from the recent reports, the efficiency of the NFA based devices is found to be steadily increased when the total energy loss decreases as shown in the Fig. 1 (b). In this context, the ML model demonstrated in this report is highly desirable for identifying the suitable materials that provide minimal energy losses in devices.

3. Methods

An ML model was implemented by collecting a set of easily accessible experimental data reported in the literatures. The present workflow consists of three major steps: (1) data collection and feature engineering, (2) model training and validation and (3) deployment of optimized model as shown in Fig. 2. The dataset including the molecular descriptors of polymers and NFAs were used for the model building.

3.1. Dataset construction

The rapid advancement in conjugated polymers and NFAs based devices leads to a large number of recent publications which offer critical information on numerous organic materials with their energy levels and different processing conditions. Approximately data set of 440 experimental observations was manually collected from the literature reported in the years 2015 to 2021 using the keyword "non-fullerene organic solar cells" in SCOPUS scientific database. This dataset consists of molecular energy levels, energy offsets, donor–acceptor ratios and processing conditions for unique combinations of polymer donors and NFAs. To implement the ML model, the collected data was randomly divided into two different datasets for training (80 %) and testing (20 %) purposes. The training and testing data sets are used for learning and performance evaluation of the ML model, respectively. A delicate balance was maintained between the training and testing datasets using the random sampling method for a stable estimation of the model performance. The performance parameters and corresponding energy loss values were manually collected from more than 160 research articles as shown in the Table S1. The effects of solvents, donor–acceptor morphology and side chains of molecules were not considered in this work. If the photophysical properties of active layers significantly altered due to any of these factors, the predicted energy loss may significantly differ from experimental energy loses. Moreover, the performance analysis of devices based on chemical structure, excited state and environment of the donor and acceptor can also be utilized for the

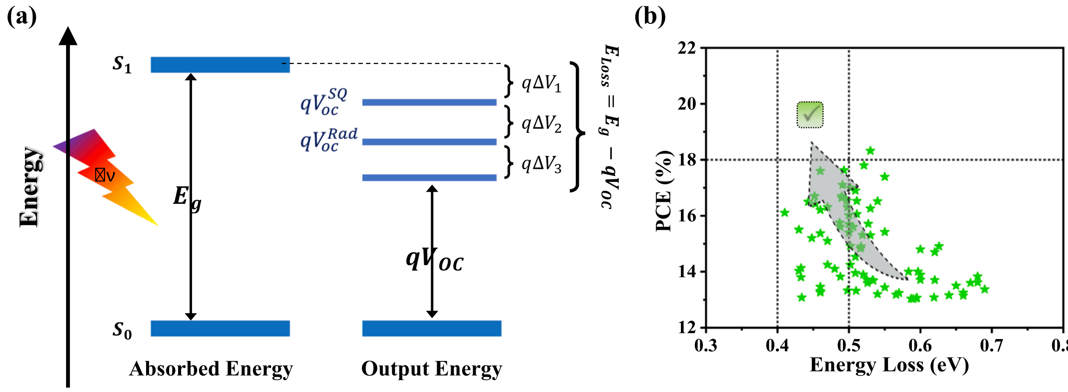

Fig. 1. (a) Schematic of the different type of losses in OSCs and (b) energy loss vs PCE for recently reported highly efficient OSCs.

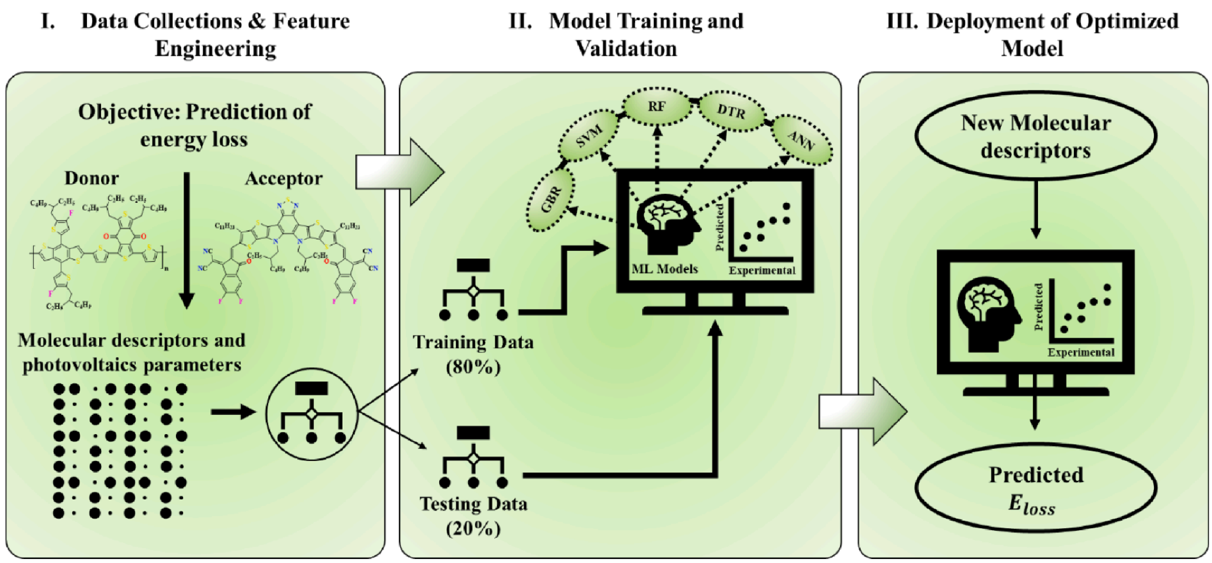

Fig. 2. The schematic of the workflow of building, application, and evaluations of ML methods.

further optimization. This will give an efficient path to improve the V_{OC} without much affecting the J_{SC} and FF.

3.2. Machine learning model and performance evaluation

The scikit-learn Python library was employed to train and test the five supervised ML regression models such as DT, SVM, GB, RF and ANN and different in-house libraries were used for the data visualization (Pedregosa et al., 2011). The details of these models were previously reported (Géron, n.d.). The performance of the model was evaluated by estimating the statistical parameters such as mean absolute percentage error (MAPE), root mean squared error (RMSE), and Pearson correlation coefficient (r) using the following equations.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{(x_i - y_i)}{x_i}\right| \times 100 \tag{3}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(y_i - x_i)^2}{n}} \tag{4}$$

$$r = \frac{\sum_{i=1}^{n}(x_i - \overline{x})\sum_{i=1}^{n}(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x})^2\sum_{i=1}^{n}(y_i - \overline{y})^2}} \tag{5}$$

where, n, x_i and y_i represent total number of data, experimental and predicted values, respectively. The terms $\overline{x}$ and $\overline{y}$ denote the average of experimental and predicted values, respectively. In general, the r value of 0.5 – 1.0 indicates a strong positive correlation between the two variables.

4. Results and discussion

4.1. Statistical analysis

To understand the reliability of collected dataset and ensure the correlation of energy loss with molecular and photovoltaic parameters, a detailed statistical analysis was carried out. The distribution histograms of PCE and energy loss are shown in Fig. 3 (a) and (b), respectively. The PCE and energy loss values were varied in the range of 0.26 – 18.32 % and 0.40 – 0.97 eV, respectively. To ensure the linear dependence of descriptors with energy loss, the Pearson coefficients were calculated and the correlation heatmap was created as shown in Fig. 3 (c). All the descriptors presented in the heatmap showed a noticeable and lesser correlation with the energy loss, indicating that no irrelevant and highly dependable parameters have been taken for performing the ML algorithms.

PCE of a solar cell shows an equal dependency on the parameters J_{SC}, V_{OC} and fill factor (FF) by considering the relation $PCE = J_{SC}.V_{OC}.FF.P_{in}^{-1}$, where P_{in} is the incident power. The joint distribution plots of PCE as functions of J_{SC}, FF and V_{OC} are shown in the Fig. 4 (a), (b) and (c), respectively. The variations of both J_{SC} (0.84 –

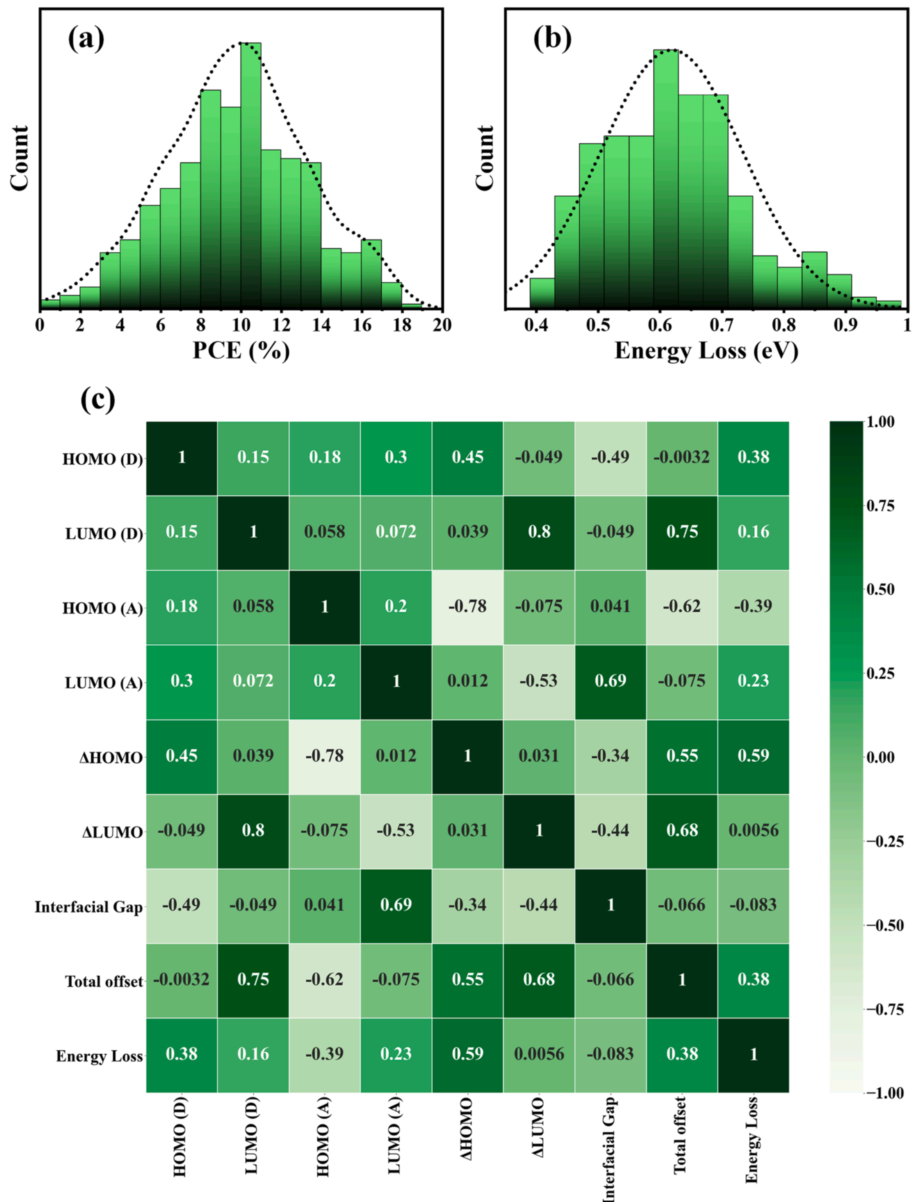

Fig. 3. Histograms of the (a) PCE (%) and (b) energy loss distributions, and (c) correlation heatmap between energy loss and material parameters.

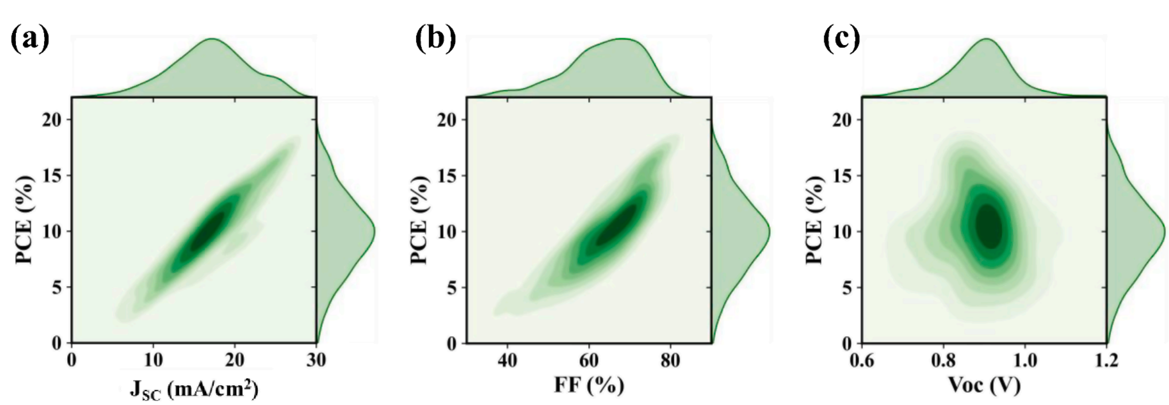

Fig. 4. Joint distribution plots for PCE with (a) J_{SC} (b) FF and (c) V_{OC} of collected data sets.

27.14 mA/cm^2) and FF (23.88 – 81.50 %) significantly changed the PCE. However, the PCE exhibited a slightly higher linear correlation with the J_{SC} than FF, indicating that J_{SC} is the most dominating factor in determining the PCE. On the other hand, the plot of PCE vs V_{OC} was more scattered with a narrow V_{OC} range of 0.7 – 1.0 V (Fig. 4 (c)), indicating that there is no notable linear correlation between PCE and V_{OC}. Previous studies indicated that further improvement in V_{OC} is limited mainly due to the occurrence of high non-radiative recombination loss (Qian et al., 2018; Wang et al., 2021; Zhang et al., 2020), which is a major contributor in the total energy loss. Therefore, minimizing the total energy loss by maintaining maximum J_{SC} and FF simultaneously is necessary to develop high performance devices.

The energy-level offset at the interface between donors and acceptors is essential for splitting excitons into free charge carriers. However, the high energy level offset causes high voltage loss limiting the performance of devices (Figure S1). In the present dataset, HOMO and LUMO levels of materials were determined using photoelectron spectroscopy or cyclic voltammetry. A statistical analysis was systematically carried out to understand the correlation between performance parameters and energy offset. Fig. 5 (a) and (b) show the correlation of J_{SC} with ΔHOMO and ΔLUMO, respectively and corresponding joint distribution plots are shown in figure S2. The J_{SC} value was found to be increased with a decrease in ΔHOMO, and the peak value was observed around 0.2 eV. J_{SC} values more than 25 mA/cm^2 were reported even at ΔHOMO ~ 0 eV (Yu et al., 2020). On the other hand, the plot of J_{SC} with ΔLUMO was more scattered with an uncertain peak at ΔLUMO = 0.2 – 0.5 eV and J_{SC} drastically decreased at ΔLUMO below 0.2 eV. Similar trend was observed in the case of variation of PCE with ΔHOMO and ΔLUMO as shown in Fig. 5 (c) and (d), respectively. PCE of more than 15 % was observed at ΔHOMO of 0 – 0.2 eV and ΔLUMO of 0.2 – 0.5 eV, and it significantly decreased at ΔLUMO below 0.2 eV. Previous reports indicate that the exciton dissociation at low energy offsets is attributed to the strong inter molecular electrostatic field induced by the electrostatic surface potential at donor–acceptor interface (Markina et al., 2021; Wang et al., 2021). The device with J_{SC} more than 17 mA/cm^2 and energy loss of 0.41 eV was demonstrated in polymer:NFA system with a negative ΔHOMO (Sun et al., 2020). A high external quantum efficiency more than 80 % was also reported even in the absence of energy offsets (figure S3). Based on above analysis and discussion, it is well understood that recent research focuses on minimizing energy offsets and energy losses to achieve higher V_{OC} leading to PCE of more than 20 %.

4.2. Performance of machine learning models

ML techniques can efficiently identify the suitable molecules to develop high performance devices and predict the structure-performance parameters and energy losses. Different ML regression models were trained in this regard and evaluated to examine their accuracy in predicting the energy loss in polymer:NFA based OSCs. The 5-fold cross-validation (CV) technique was used in this study for the training and testing of ML models, and the obtained parameters from each CV round were averaged to ensure the realistic performance of the ML models. The correlations between the experimental and predicted energy losses obtained by performing the ML models such as DT, SVM, GB and RF are shown in Fig. 6 (a) – (d), respectively. To compare their predictive power, the statistical parameters were calculated and summarized in the Table 1. All the models possess high correlation coefficients r more than 0.70 between experimental and predicted energy losses. The RF regression model was found to be more efficient compared to other models for energy loss predictions. This model showed a strong linear relationship between the experimental and predicted energy losses with MAPE of 7.093 %, RMSE of 0.063 eV and r of

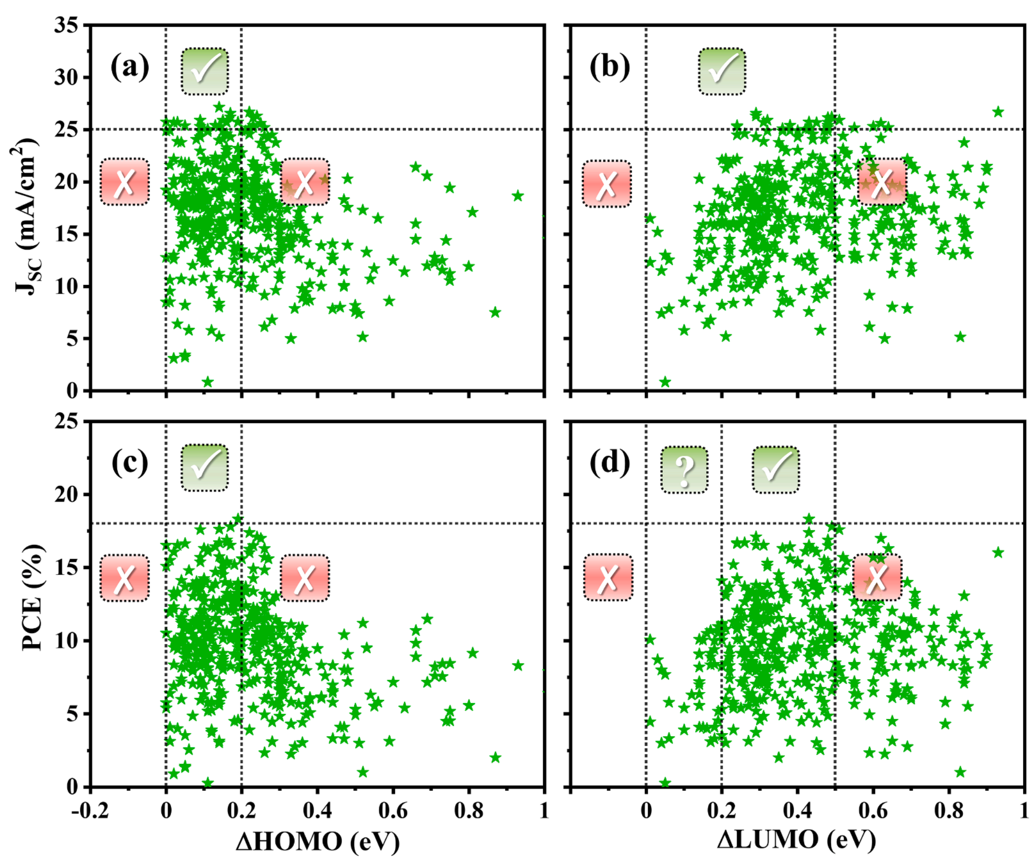

Fig. 5. Correlation plots for (a) J_{SC} vs ΔHOMO, (b) J_{SC} vs ΔLUMO (c) PCE vs ΔHOMO, and (d) PCE vs ΔLUMO. ΔHOMO and ΔLUMO represent the difference between HOMO and LUMO of donor and acceptor as shown in figure S1.

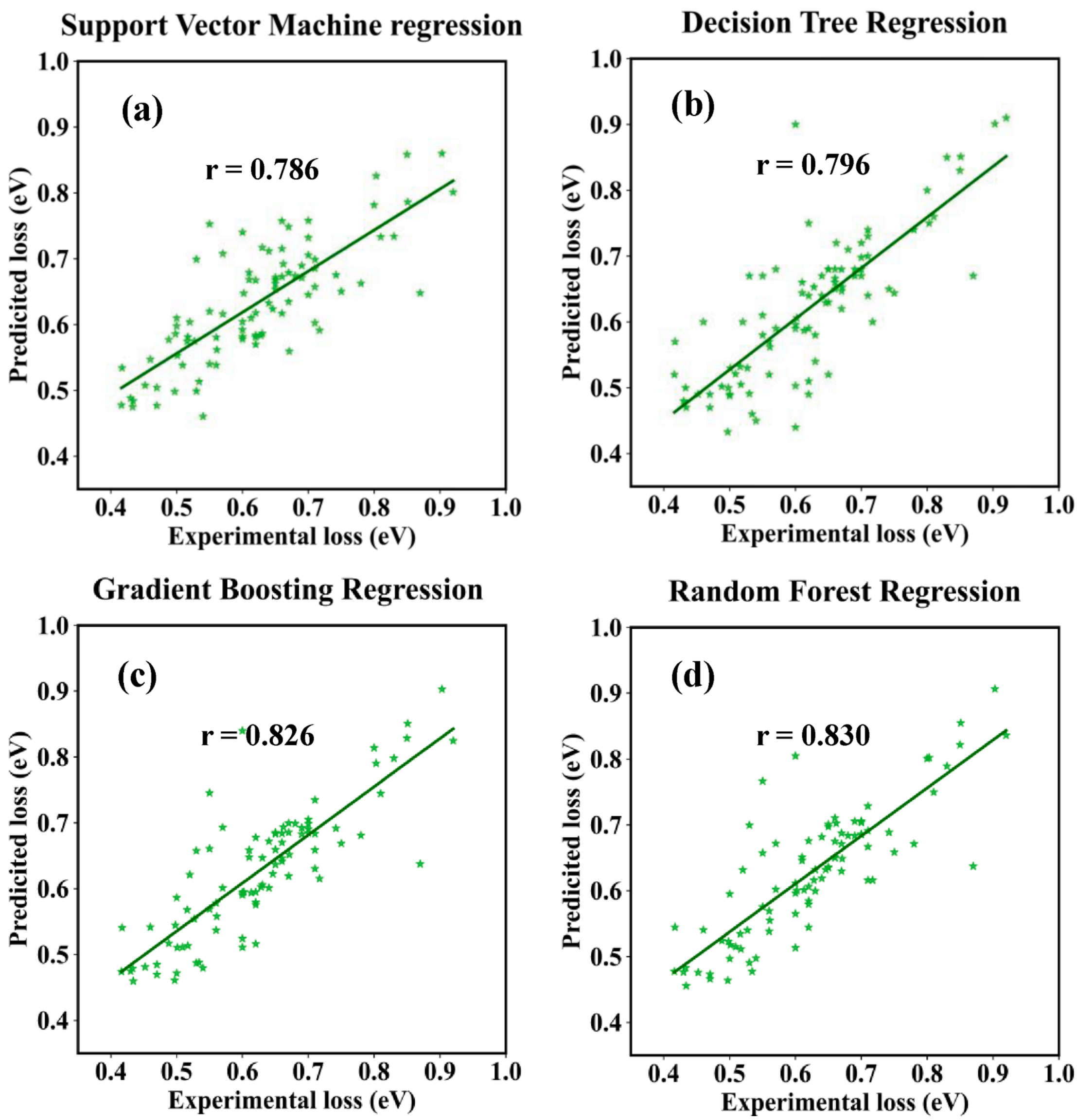

Fig. 6. Correlation between the experimental and predicted energy loss for (a) SVM, (b) DT, (c) GB and (d) RF models.

Table 1
Performance evaluation parameter for different ML model.

ML Model	MAPE [%]	RMSE (eV)	r
Support Vector Machine Regression	9.060	0.071	0.787
Decision Tree Regression	8.186	0.071	0.797
Gradient Boosting Regression	7.488	0.064	0.826
Random Forest Regression	**7.093**	**0.063**	**0.830**
Artificial Neural Network	8.561	0.072	0.778

0.830 as it randomly fits many decision trees to enhance the performance. The small value of RMSE and the high value of r indicate the superior performance of RF regression model. ANN model was also performed on the present datasets as it generally exhibits high predictive power in the classification problems. However, this model showed slightly declined performance with MAPE of 8.561 %, RMSE of 0.072 eV and r of 0.778. Generally, the ANN model requires very large amount of data to train neural networks for understanding complex relations between the descriptors. The relative error in the prediction of energy losses was found to be in the range of 0 – 20 % for the testing dataset indicating that present RF regression model is flexible enough to provide the desired accuracy to predict the energy loss in the polymer:NFA based OSCs (Figure S4). The present ML model exhibited similar performance with previously reported models employed for the prediction of PCE and V_{OC} in OSCs (Lee, 2020; Malhotra et al., 2021; Sahu et al., 2018).

To further investigate the predictive ability, the present RF regression model was employed on the devices with different combinations of polymer donors and NFAs. OSCs based on D-A conjugate polymers and fused ring-based acceptors such as PBDB-T, PM6, Y6, ITIC etc. with wide PCE range of 6.30 – 17.80 % were utilized from the previous reported literature to verify this model. The energy loss values were predicted and compared with the experimental values as shown in the Table 2. This model provided significantly closer experimental and predicted energy loss values. For example, the devices based on the promising donor–acceptor combinations PM6:Y6, PBDB-TF:BTP-eC9 provided relative errors of 2.82 % and 1.15 %, respectively. Based on the above investigations, we strongly believe that the ML is one of the suitable approaches to identify the novel materials with optimal molecular descriptors to achieve high performance devices with minimal energy losses. In general, this approach can provide an effective prediction model and analysis of the importance of various input features based on experimental data, avoiding the time-consuming high throughput experiment.

Table 2
Predicted and experimental energy loss value based on the RF model.

Donors	Acceptors	PCE (%)	Experimental loss (eV)	Predicted loss (eV)	Relative Error (%)	Ref.
P3HT	o-IDTBR	6.30	0.910	0.850	6.10	(Holliday et al., 2016)
PBDB-T	ITIC	10.40	0.700	0.720	2.71	(Liu et al., 2018)
J71	IT-4F	11.66	0.730	0.720	2.98	(Hao et al., 2019)
PM6	Y15	14.13	0.433	0.485	12.05	(Luo et al., 2019)
PM6	Y6	15.70	0.500	0.490	2.82	(Yuan et al., 2019)
PBDB-TF	BTP-eC9	17.80	0.520	0.510	1.15	(Cui et al., 2020)

4.3. Experimental validations

To validate the effectiveness of our ML model, the real photovoltaic devices were fabricated by selecting three different D/A combinations of active layers. These combinations were specifically chosen as they have optimized HOMO and LUMO offsets, making them suitable for developing high-performance devices with PCE over 10 %. The chemical structures of donors and acceptors used for the device fabrication are shown in Fig. 7 (a), the current density–voltage (J-V) curves of the devices are presented in Fig. 7 (b), and their photovoltaic parameters with energy loss values are summarized in Table 3. The devices fabrication procedure are discussed in the supplementary information. The devices based on PBDB-T:ITIC, PM6:Y6, and PM6:BTP-eC9 active layers exhibited PCE of 9.97 %, 13.48 %, and 14.68 %, respectively. The experimental energy loss values for PBDB-T:ITIC, PM6:Y6 and PM6:BTP-eC9 active layers were found to be around 0.709 eV, 0.492 eV, and 0.520 eV, respectively, which are very close to the values obtained from RF model predictions. Therefore, the present results also confirmed the high accuracy and superiority of the RF model in energy loss predictions. The outcome of this study with experimental verifications reveals that the present ML approach can open up a new perspective in the designing of novel materials and the appropriate selection of D/A pairs for accelerating the development of high-performance OSCs.

Table 3
The photovoltaic performance parameters of OSCs with different D/A combinations under AM 1.5 G illumination (100 mW cm^{-2}).

Active layers	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)	Experimental loss (eV)	Predicted loss (eV)
PBDB-T:ITIC (Eg = 1.59 eV)	0.881	18.37	61.62	9.97	0.709	0.720
PM6:Y6 (Eg = 1.33 eV)	0.838	24.37	66.02	13.48	0.492	0.490
PM6:BTP-eC9 (Eg = 1.36 eV)	0.840	27.15	64.39	14.68	0.520	0.510

5. Conclusion

In summary, a statistical analysis was carried out on 440 data collected from 162 research articles to understand the recent research progress in polymer:NFA based OSCs. The performance of the devices steadily increased when the total energy loss decreased. High J_{SC} and PCE more than 25 mA/cm^2 and 18 % were reported, respectively at ΔHOMO of 0–0.2 eV and ΔLUMO of 0.2 – 0.4 eV. The statistical studies indicated that the energy loss and energy level offsets are the most critical parameters in determining the performance. In addition, the structure-performance relation of devices was systematically studied by employing ML models using various molecular descriptors such as energy level offsets, band gap, etc. Different ML regression models such as SVM, DT, GB, RF and ANN were used to predict the energy losses. Among these models, the RF model was found to be efficient, which exhibited a correlation coefficient of r = 0.830 with MAPE of 7.093 % and RMSE of 0.063 eV. The devices with selected D/A combinations exhibited closely matching predicted and experimental energy loss values, which further confirmed the predictive ability of the RF model. Our findings indicate that the present RF model is suitable to predict the energy losses, which may provide vital information for further improvement in device performance. We hope similar works will substantially aid the development of OSCs by improving our understanding

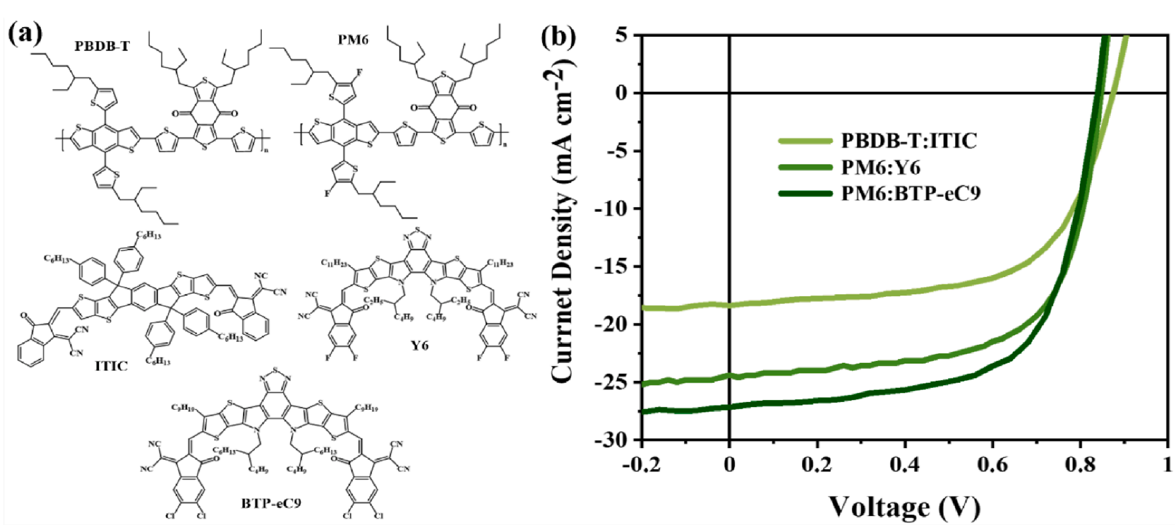

Fig. 7. (a) Chemical structures of donors (PBDB-T and PM6) and the acceptors (ITIC, Y6 and BTP-eC9), and (b) the J-V characteristics of the fabricated devices.

of device working mechanisms and design rules for new active layer materials.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.solener.2022.12.029.

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